

Dialysis is a process for removing waste and excess water from the blood and is used primarily as an artificial replacement for lost kidney function in people with kidney failure. Dialysis filters out unwanted substances and fluids from the blood.

3. The average blood flow through kidneys per minute is –

- (a) 1000 cc (b) 1200 cc
(c) 200 cc (d) 500 cc

56th to 59th B.P.S.C. (Pre) 2015

Ans. (b)

Kidneys are the most important organ in our body. Kidneys help in purification of blood and also removes toxic materials from our body through urine. Our kidneys purify around 1500 litre of blood and excrete approximately 1.5 litre urine per day. 1200 ml (one ml equal to one cubic cm.) of blood flows through both the kidneys per minute and out of it 1 ml of urine is formed per minute.

4. The element excreted through human sweat is :

- (a) Sulfur (b) Iron
(c) Magnesium (d) Zinc
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

Sweat is produced by the skin in the form of liquid to regulate the body temperature. It is the part of the excretory function of the skin. Sodium, potassium, calcium, magnesium and many other trace elements (e.g. zinc, copper, iron, chromium, nickel and lead) are excreted through human sweat.

VI. Nervous System

1. Input and output nerves meet at

- (a) liver (b) central nervous system
(c) heart (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (b)

The Central Nervous system has four parts–(1) Brain, (2) Spinal cord, (3) Nerve, (4) Neuron. The input and output nerves meet at several places particularly in the spinal cord.

2. How many pairs of nerves arise from the spinal cord?

- (a) 13 (b) 31
(c) 33 (d) 12

45th B.P.S.C. (Pre) 2001

Ans. (b)

Nerves that arise from the spinal cord are called spinal nerves. A spinal nerve is a mixed nerve, which carries motor, sensory and autonomic signals between the spinal cord and the body. In human there are 31 pairs of spinal nerves, one on each side of the vertebral column. These are grouped into the corresponding cervical, thoracic, lumbar, sacral and coccygeal regions of the spine. There are eight pairs of cervical nerves, twelve pairs of thoracic nerves, five pairs of lumbar nerves, five pairs of sacral nerves and one pair of coccygeal nerves. The spinal nerves are part of the peripheral nervous system.

3. The activities like walking in a straight line, riding a bicycle and picking up a pencil are possible due to which part of the brain?

- (a) Hypothalamus (b) Cerebellum
(c) Cerebrum (d) Midbrain

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

The activities like walking in a straight line, riding a bicycle and picking up a pencil are possible due to cerebellum part of the Brain. The cerebellum is located at the back of the Brain, immediately inferior to the occipital and temporal lobes and within the Posterior Cranial Fossa.

4. Where is the governing centre of reflex action located?

- (a) In cerebrum (b) In cerebellum
(c) In spinal cord (d) In nerve cells

45th B.P.S.C. (Pre) 2001

Ans. (c)

A reflex arc is a neural pathway that controls an action reflex. In higher animals, most sensory neurons do not pass directly into the brain, but synapse in the spinal cord. This characteristic allows reflex actions to occur relatively quickly by activating spinal motor neurons without the delay of routing signals through the brain, although the brain will receive sensory input while the reflex action occurs. There are two types of reflex arc- autonomic reflex arc (affecting inner organs) and somatic reflex arc (affecting muscles).

5. Which parts of the brain control blood pressure?

- (a) Spinal cord, skull, hypothalamus
(b) Spinal cord, skull, cerebrum
(c) Pons, medulla, cerebellum
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

Pons, Medulla and Cerebellum of the brain control the blood pressure.

6. Body temperature is regulated by :

- (a) Thalamus (b) Hypothalamus
(c) Cerebellum (d) Medulla
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

64th B.P.S.C. (Pre) 2018

Ans. (b)

The average normal temperature of the human body is 98.6° F. The hypothalamus controls body temperature. It also controls hunger, important aspects of parenting and attachment behaviours, thirst, fatigue, sleep and circadian rhythms.

7. Birds excrete nitrogenous waste in the form of

- (a) urea (b) ammonia
(c) fatty acid (d) uric acid
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

Birds excrete nitrogenous waste in the form of uric acid. Birds excrete through a body part called the cloaca. Birds excrete nitrogenous wastes as uric acid in the form of a white pellet or paste with minimum loss of water and are called as uricotelic animals.

8. Which one of the following organs most actively functions to regulate our body temperature?

- (a) Skin (b) Heart
(c) Lungs (d) Stomach
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

Thermoregulation is the process by which the human body keeps its temperature regulated. Skin plays a vital and indispensable role in maintaining body temperature. It offers a barrier between the internal structure of the body and the outside environment.

9. Pituitary gland is located in-

- (a) Intestine (b) Liver
(c) Kidney (d) Brain
(e) None of the above /More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

The pituitary gland is a small endocrine gland located below the hypothalamus in the brain.

10. What is the function of the pituitary gland?

- (a) To develop sex organs in males
(b) To stimulate growth in all organs

- (c) To regulate sugar and salt levels in the body
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (d)

The pituitary gland also known as the master gland. Function of pituitary gland is as follow–

1. To develop sex organs in males,
2. To stimulate growth in all organs,
3. To regulate salt and sugar level in the body.

11. Which of the following is not an endocrine gland?

- (a) Thyroid (b) Pituitary
(c) Lacrimal (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (c)

Endocrine glands lack ducts and are hence, called ductless glands. Their secretions are called hormones. The classical definition of hormone is a chemical produced by endocrine glands and released into the blood and transported to a distantly located target organ.

The endocrine glands and hormone producing diffused tissues/cells located in different parts of our body constitute the endocrine system. Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis in males and ovary in females) are the organised endocrine bodies in our body.

12. Which one of the following statements is incorrect about muscles in human body?

- (a) Muscles can only push the bone.
(b) Muscles work in pairs.
(c) Muscles become shorter, stiffer and thicker when contracted.
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Muscles can pull bones but they cannot push them back to the original position. So they work in pairs of Flexors and extensors. So the option (a) is an incorrect option. Muscles is a specialised tissue of Mesodermal origin. About 40-50 percent of the body weight of a human adult is contributed by muscles. They have special properties like excitability, contractibility, extensibility and elasticity. Based on their location, three types of muscles are identified–

1. Skeletal Muscles
2. Visceral Muscles
3. Cardiac Muscles